



- 1 A thin cylindrical bar magnet of length L and cross-sectional area A is attached to a block. An identical magnet is attached to a trolley, as shown in Fig. 1.1.

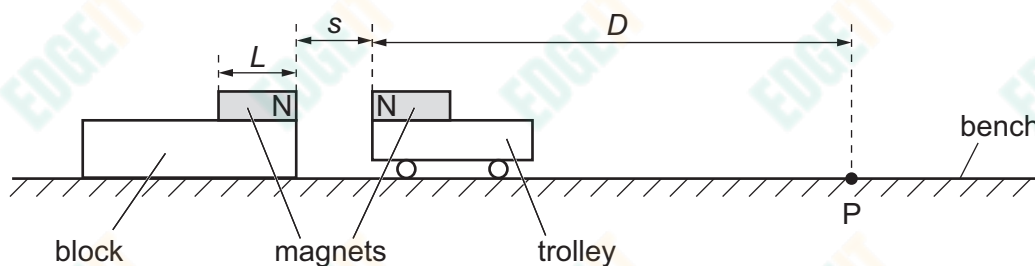


Fig. 1.1

The trolley is held so that the separation of the N poles of the two magnets is s .

Point P is a distance D from the N pole of the magnet on the stationary trolley.

The trolley is released. The speed v of the trolley at point P is determined using one light gate.

It is suggested that v is related to s by the relationship

$$\frac{mv^2}{2D} = \frac{KA^2B^2L^2}{s^4} - Q$$

where B is the magnetic flux density at the N pole of one of the magnets, m is the mass of the trolley, and K and Q are constants.

Plan a laboratory experiment to test the relationship between v and s .

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and Q .

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.





Diagram

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- 2 A student investigates an electrical circuit. A power supply of electromotive force (e.m.f.) E_s and negligible internal resistance is connected in series to three resistors, each of resistance Z .

A cell, an ammeter and a resistor of resistance R are connected in parallel across one of these resistors, as shown in Fig. 2.1.

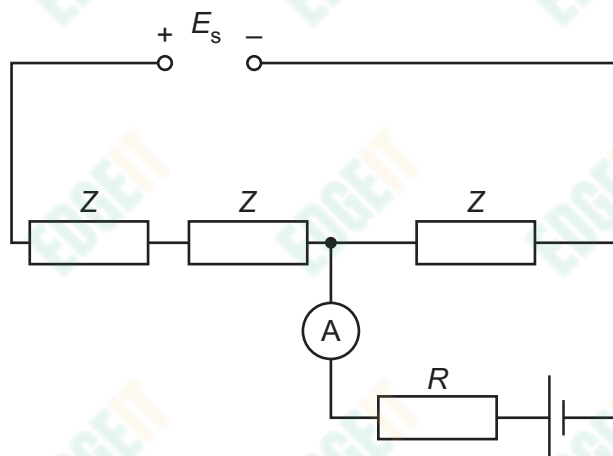


Fig. 2.1

The current I is measured by the ammeter for different values of R .

It is suggested that I and R are related by the equation

$$3E - E_s = I(3R + 2Z)$$

where E is the e.m.f. of the cell.

- (a) A graph is plotted of $\frac{1}{I}$ on the y -axis against R on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]





(b) Values of R and I are given in Table 2.1.

Table 2.1

$R/\text{k}\Omega$	$I/\mu\text{A}$	$\frac{1}{I}/\text{A}^{-1}$
1.50	194 ± 2	
1.75	180 ± 2	
1.92	172 ± 2	
2.22	160 ± 2	
2.48	150 ± 2	
2.72	144 ± 2	

Calculate and record values of $\frac{1}{I}/\text{A}^{-1}$ in Table 2.1.

Include the absolute uncertainties in $\frac{1}{I}$.

[2]

(c) (i) Plot a graph of $\frac{1}{I}/\text{A}^{-1}$ against $R/\text{k}\Omega$. Include error bars for $\frac{1}{I}$.

[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

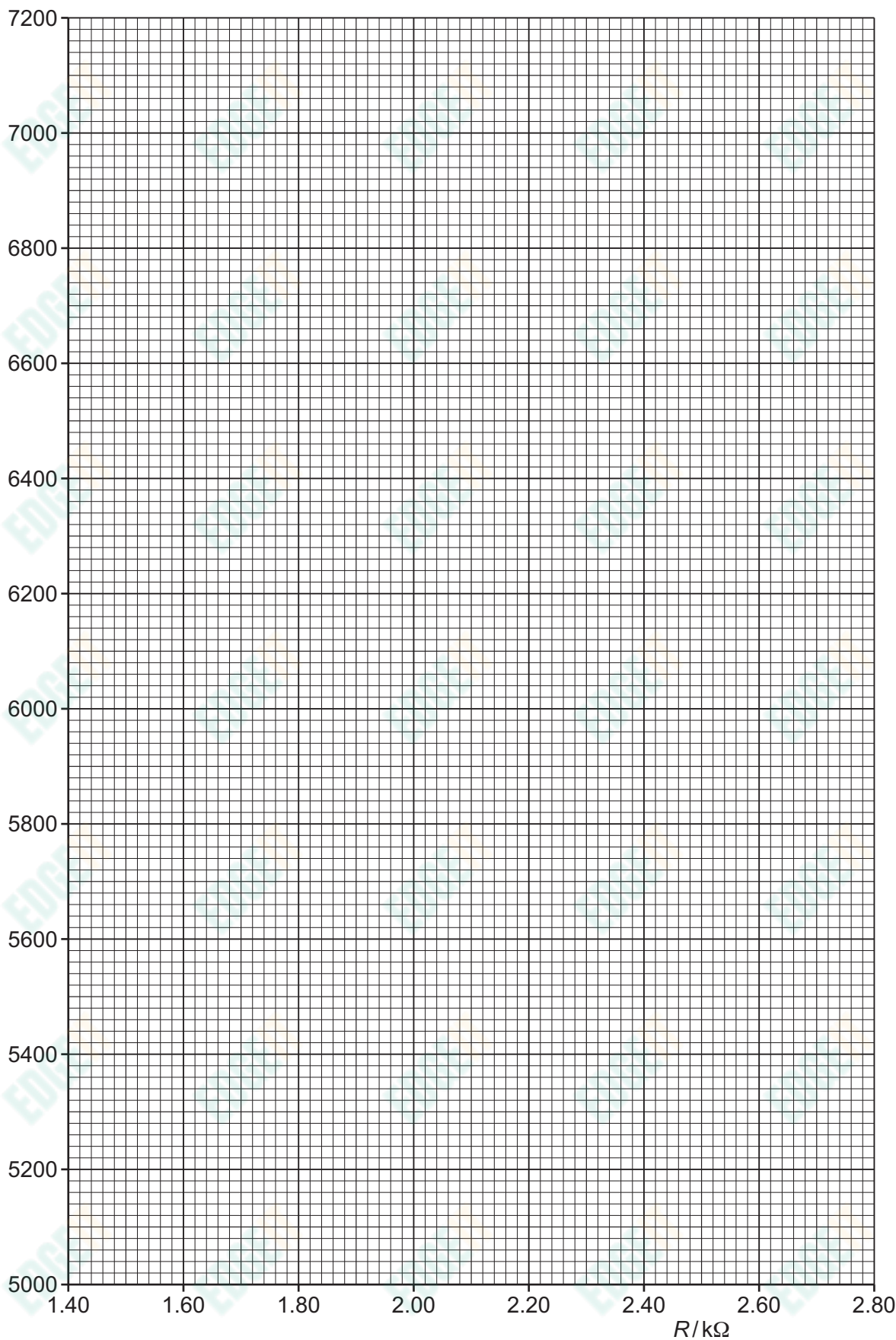
(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]





$\frac{1}{I} / \text{A}^{-1}$





- (iv) Determine the y -intercept of the line of best fit. Include the absolute uncertainty in your answer.

y -intercept = [2]

- (d) (i) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of E and Z . Include appropriate units.

Data: $E_s = (2.20 \pm 0.05)V$

$E =$

$Z =$ [2]

- (ii) Determine the absolute uncertainty in E .

absolute uncertainty in $E =$ [1]

- (e) The experiment is repeated. Determine the resistance R that gives a value of I of $250\mu A$.

$R =$ Ω [1]

[Total: 15]

